

AMAN KUMAR SINGH

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Summary

Aspiring AI Engineer with expertise in Python, Machine Learning, Data Analytics, Generative AI, and Backend Development. Experienced in building LLM-based applications using FastAPI, React, Node.js, LangChain, RAG, and modern AI frameworks through academic projects and internships. Seeking opportunities in AI Engineering, Generative AI, or Data Science to build intelligent, scalable solutions.

Education

Master of Computer Application

Sharda University, Uttar Pradesh

PG: 2026

CGPA: 9.27 / 10

Bachelor of Computer Application

JIS College of Engineering, West Bengal

UG: 2024

CGPA: 8.14 / 10

Experience

Healthletic Lifestyle

Backend Engineering Intern

Remote

Jun 2026 – Present

- Developed and maintained scalable **RESTful APIs** using **Node.js**, **Express.js**, and **MongoDB**, implementing business logic, database operations, **JWT authentication**, and request validation to deliver secure and reliable backend services.
- Tested, validated, and documented APIs using **Swagger** and **Postman** across local and production environments, integrated third-party services, and ensured seamless communication between frontend and backend systems.
- Collaborated with cross-functional teams to identify and resolve backend bugs, implement new features, support deployment, and manage source code using **Git** while following industry-standard software development practices.

Projects

ResumeIQ–AI : Generative AI-Powered Resume Analyzer | React.js, Node.js, Groq, Gen AI

Jun 2026



- Developed a full-stack **Generative AI-powered Resume Analyzer** using **React.js**, **TypeScript**, **Node.js**, **Express.js**, **MongoDB**, and **Groq (Llama 3.3)**, enabling ATS score analysis, contextual resume insights, and AI-driven career assistance.
- Engineered a secure resume processing pipeline using **Multer**, **PDF parsing**, **Cloudinary**, **JWT Authentication**, **bcrypt**, and **Nodemailer** to support PDF upload, cloud storage, live preview, protected APIs, and email verification.
- Built an intelligent **LLM-powered Resume Chat Assistant** with modular RESTful APIs, providing resume-specific Q&A, ATS optimization, personalized career recommendations, and a responsive user experience using **Tailwind CSS** and **shadcn/ui**.

Heart-AI: Dual-Engine Cardiovascular Diagnostic | Python, Flask, TensorFlow, XGBoost

Mar 2026



- Engineered a dual-engine cardiovascular diagnostic platform utilizing **Flask**, **XGBoost**, and **CNN-based deep learning** to concurrently analyze tabular clinical vitals and physical ECG image scans.
- Trained and deployed high-accuracy machine learning models on complex medical datasets, achieving a **92%** prediction accuracy for real-time heart disease risk assessment and classification.
- Designed a complete full-stack analytics dashboard integrating **SQLite**, **Chart.js**, and **FPDF** to deliver interactive health visualizations, patient management archiving, and automated hospital-grade PDF reports.

Air Quality Analysis & Forecasting Dashboard | Python, Flask, LSTM

Jan 2026



- Developed a web-based air quality monitoring system integrating real-time environmental data using APIs to track AQI and pollutant levels (PM2.5, PM10, NO₂, CO, O₃).
- Performed data cleaning and exploratory data analysis (EDA) using **Pandas** and **NumPy** to clean, transform, and analyze air quality datasets.
- Built a **LSTM-based deep learning model** to forecast PM2.5 levels for the next 24 hours, enabling predictive analysis of pollution trends.

Technical Skills

- Languages:** Python, Java, JavaScript, TypeScript, SQL
- Data Analytics:** Pandas, NumPy, Power BI, Excel, Data Visualization, EDA
- AI & ML:** Machine Learning, Deep Learning, Scikit-learn, TensorFlow, Time Series Forecasting
- Generative AI:** LangChain, RAG, Hugging Face, ChromaDB, Vector DB, Prompt Engineering, LLM Applications
- Backend Development:** FastAPI, Flask, Node.js, Express.js, REST APIs, Spring Boot
- Databases:** MySQL, PostgreSQL, MongoDB, SQLite
- Tools & Platforms:** Git, GitHub, Postman, VS Code, Jupyter Notebook

Certifications

- Data Visualization using Power BI:** Sharda Informatics — May–June 2025 
- Prompt Engineering:** Sharda Informatics — Jan–April 2025 
- AI: Transformative Learning:** TechSaksham — 2025 
- Python Pro Bootcamp:** Udemy — 2024 
- Mastering in C:** Udemy — 2022 